DATA SCIENCE | DS-A3 | HOMEWORK

Predicting adult Red Knot sex from morphometric data

Regression Baseline and Diagnostics

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Research question

How accurately can the sex of adult Red Knots (*Calidris canutus canutus*) be predicted from wing, bill and tarsus measurements, and which individual measurements or combinations perform best?

Data and analytical sample

The source is the adult Mauritanian dataset accompanying Oortwijn et al. (2025), *Demand-resource mismatch explains body shrinkage in a migratory shorebird* [1]. Birds were captured at Banc d'Arguin. Reference sex was determined by DNA analysis. Wing, bill and diagonal tarsus lengths are in millimetres.

Preparation step	Records / individuals
Original file	2,165 records; 2,131 ring identifiers
Missing values / identical rows	0 / 0
Conflicting reference sex	Both records for Z062234 excluded
Repeated captures	33 later records excluded after conflict removal
Analytical sample	2,130 birds: 1,263 females and 867 males

One earliest record per ring identifier was retained, ordered by Year and Month. Selection did not depend on morphometric values. The original data remain unchanged; excluded records are retained separately in the notebook session.

Frozen split and evaluation design

Sex	Training	Test
Female	1,010	253
Male	694	173
Total	1,704	426

An 80:20 split was stratified by sex with random_state=42. Ring identifiers do not overlap between subsets. The split assignments are saved in `split_manifest.csv`.

All candidate comparisons use five stratified cross-validation folds within the training set (shuffle=True; random_state=42). StandardScaler is inside each Pipeline. Balanced accuracy is the primary metric; ordinary accuracy is secondary. The majority baseline always predicts the most frequent fitting-set sex.

2. Candidate models and complexity gains

Seven unregularized logistic models cover every non-empty subset of Wing, Bill and Tarsus. The solver is lbfgs, max_iter=1000, with no coefficient penalty. Ridge (L2) and lasso (L1) are examined for Wing + Bill and all three measurements.

For each regularized candidate, three-fold stratified inner cross-validation selects C by balanced accuracy from {0.001, 0.01, 0.1, 1, 10, 100, 1000}. Smaller C imposes stronger regularization. Both use saga, max_iter=10000 and random_state=42. The existing five outer folds evaluate the tuning procedure.

Candidate	Predictors	Accuracy (%)	Balanced accuracy (%)
Unregularized	Wing + Bill	77.35 +/- 1.79	75.86 +/- 2.27
Unregularized	Wing + Bill + Tarsus	77.29 +/- 1.62	75.81 +/- 2.03
Unregularized	Wing	73.53 +/- 1.00	72.26 +/- 1.26
Unregularized	Wing + Tarsus	73.65 +/- 1.32	71.64 +/- 1.59
Unregularized	Bill + Tarsus	71.65 +/- 1.82	69.55 +/- 2.05
Unregularized	Bill	71.36 +/- 1.77	69.26 +/- 2.13
Unregularized	Tarsus	61.74 +/- 0.99	55.96 +/- 1.07
Majority baseline	None	59.27 +/- 0.08	50.00 +/- 0.00
Lasso	Wing + Bill + Tarsus	77.35 +/- 1.56	75.86 +/- 1.98
Lasso	Wing + Bill	77.35 +/- 1.95	75.84 +/- 2.41
Ridge	Wing + Bill + Tarsus	77.29 +/- 1.62	75.81 +/- 2.03
Ridge	Wing + Bill	77.29 +/- 1.84	75.79 +/- 2.32

Values are cross-validation means +/- sample SD across five outer folds. Means are percentages; SDs are percentage points. For regularized models, these are outer-fold scores after inner-fold tuning.

Selection and gain from complexity

Wing is the strongest individual predictor (72.26% balanced accuracy). Adding Bill raises this to 75.86%, a gain of about 3.6 percentage points. Adding Tarsus does not improve either mean score. Regularized variants achieve 75.79-75.86% balanced accuracy, with no material observed advantage. The **unregularized Wing + Bill model** is retained for its similar performance and simpler specification. Tiny score differences do not establish superiority.

Selected coefficients

Predictor	Per SD	Per millimetre
Wing	1.1336	0.2458
Bill	1.0583	0.5533

For the selected fit, F=1 and M=0. Both coefficients are positive: holding the other measurement constant, a larger measurement is associated with higher predicted female probability. Coefficients operate on the log-odds scale, not as percentage changes in probability. Similar standardized coefficients do not by themselves establish equal biological importance.

3. Residuals and classification errors

The selected model is examined using predictions from the same five cross-validation splits. Each bird is predicted by a model fitted without that bird. Female probability greater than 0.5 yields a female classification.

Reference sex	Birds	Correct	Errors	Recall (%)
Male	694	471	223	67.87
Female	1,010	847	163	83.86

Male error rate is 32.1%, compared with 16.1% for females. Overall accuracy therefore masks poorer identification of males. Pooled cross-validation accuracy is 77.35% and balanced accuracy is 75.86%.

Residual structure and heteroscedasticity

Raw residuals are y-p, with F=1 and M=0. Their two branches, 1-p for females and -p for males, follow directly from the binary response and are not by themselves evidence of model adequacy. Five fixed probability intervals (width 0.2) are used for a descriptive check.

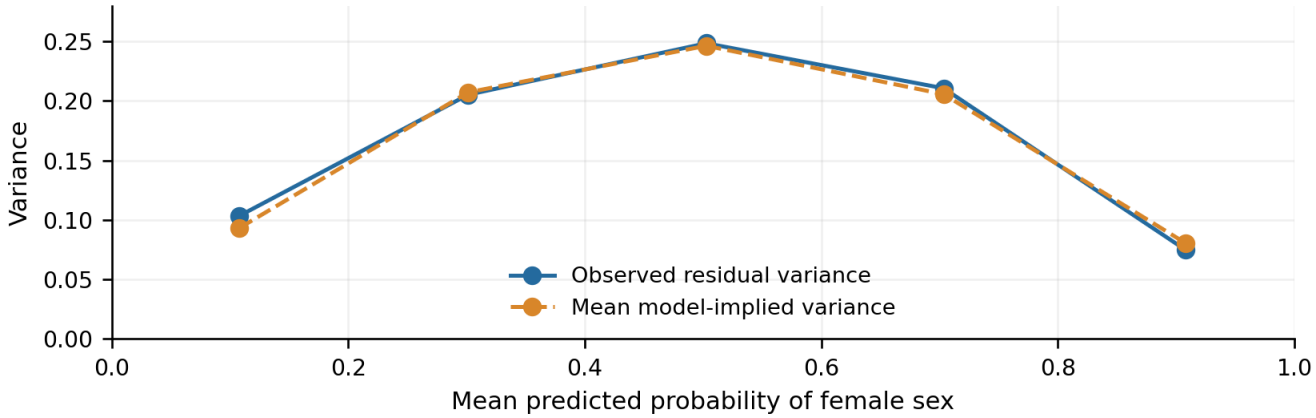


Figure 1. Observed residual variance and the mean Bernoulli reference p(1-p), calculated within each probability interval.

Mean residuals range from -0.0165 to 0.0089. Predicted and observed female fractions are close at this grouping level, which does not establish calibration for every individual or subgroup. Observed variance peaks at 0.2484 near p=0.5, close to the model-implied 0.2463. Nonconstant variance is expected for a Bernoulli response and is broadly consistent with this diagnostic.

Five confident error cases

Ring	Reference sex	Wing	Bill	P(female)	Prediction
Z062396	F	160	30.5	0.0058	M
Z049413	M	178	38.9	0.9827	F
Z063154	M	178	38.7	0.9807	F
Z023767	F	163	32.0	0.0261	M
Z049322	F	157	35.1	0.0409	M

For example, female Z062396 is assigned P(female)=0.0058, while male Z049413 is assigned 0.9827. Grouped agreement can coexist with confident individual errors. These observations remain in the analysis.

4. Influence and learning curves

Sensitivity to individual training observations

Each of the 1,704 training birds is temporarily omitted in turn and the selected model is refitted. Its predictions are compared with the full-training fit on the same training feature values. These changes measure sensitivity, not generalization accuracy. The main analytical sample is not changed.

Omitted ring	Sex	Maximum change (pp)	Mean change (pp)	Changed classes
Z049293	F	1.2586	0.1933	8
Z063209	M	1.0550	0.1277	0
Z049322	F	1.0303	0.1816	7
Z083329	F	0.9807	0.1214	3
Z101069	F	0.9263	0.1345	8

The largest probability change is 1.26 percentage points after omitting Z049293; its mean absolute change is 0.19 percentage points. Eight classifications change, which does not necessarily mean eight errors are corrected.

Coefficient	Full-training value	Largest absolute change
Wing	0.245824	0.004139
Bill	0.553310	0.007769

Maximum changes are about 1.7% of the Wing coefficient and 1.4% of the Bill coefficient. Signs remain positive. This analysis indicates limited sensitivity to removal of a single training observation.

Learning curves

The same outer folds are evaluated at five training sizes: 136, 340, 681, 1,022 and 1,363 birds per fitting subset. Training subsets are shuffled with random_state=42, and preprocessing is refitted within each subset.

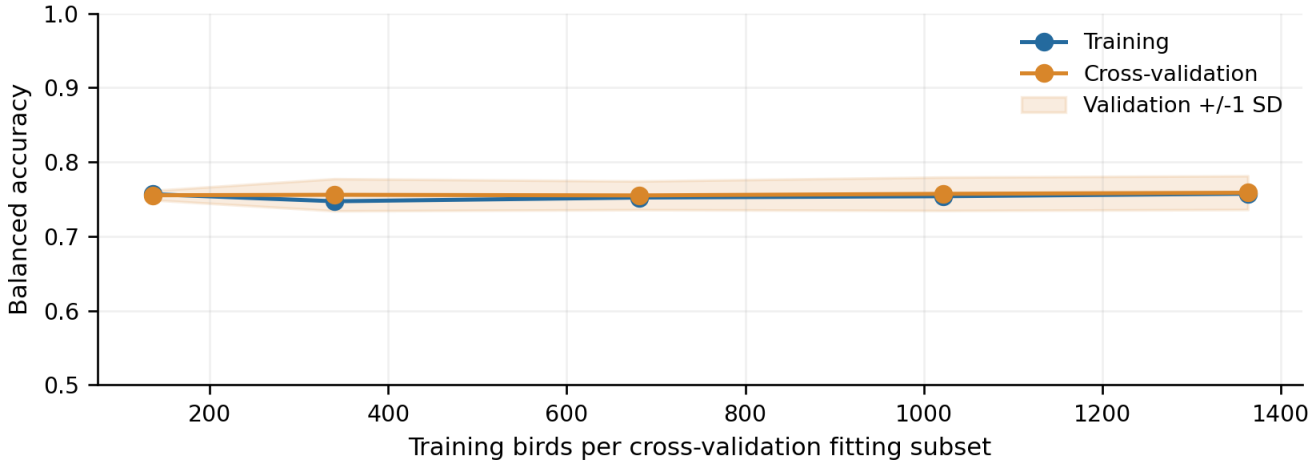


Figure 2. Training and validation balanced accuracy. Shading is +/-1 sample SD across validation folds, not a confidence interval.

Validation means range from 75.50% to 75.86%. The increase from the smallest to largest training size is about 0.34 percentage points. Training and validation scores are close, with no large gap suggesting substantial overfitting in this diagnostic. The curve is relatively flat over the examined range; this does not establish that larger datasets could never help.

5. Final test, interpretation and limits

The selected model and preprocessing were fitted to the 1,704 training birds before test evaluation. No feature selection, threshold tuning or regularization tuning uses the test outcomes. The majority baseline is also fitted using training data only.

Model	Test accuracy (%)	Test balanced accuracy (%)
Majority baseline	59.39	50.00
Wing + Bill	73.71	71.83

Reference sex	Predicted M	Predicted F	Birds	Recall (%)
Male	107	66	173	61.85
Female	46	207	253	81.82

The selected model correctly classifies **314 of 426 test birds** (73.7% accuracy; 71.8% balanced accuracy), exceeding the baseline by 14.32 and 21.83 percentage points, respectively. However, 66 males and 46 females are misclassified. Test scores are lower than the cross-validation estimates, and male recall remains substantially lower than female recall.

Why coefficients are not automatically causal

The coefficients describe conditional statistical associations in observational data. They do not estimate the causal effect of changing wing or bill length: a positive coefficient does not mean that increasing a measurement would change a bird's sex. Values also depend on the included predictors and their relationships. They are components of a predictive rule, not automatic measures of causal influence or biological importance.

Claim boundary

Wing and bill length provide useful but incomplete information about sex in this adult Red Knot dataset. Results do not support near-certain individual sex assignment, particularly for males. Additional predictors and regularization did not materially improve the observed cross-validation scores.

The random individual split evaluates birds from the same sampling framework. It does not establish transferability to juveniles, other populations or subspecies, or new locations and sampling periods. The final claim is moderate predictive performance with explicit limits, rather than universally accurate morphometric sex determination.

Computational source and verification

Source notebook: `DS_A3_red_knot_sex.ipynb`. The recorded Colab scikit-learn version is 1.6.1. An independent AI-assisted review executed all 15 code cells in sequence using scikit-learn 1.8.0, adapting only file paths; substantive numerical tables matched. The author also ran the analysis blocks in Colab and compared their outputs.

Input file: `14._adults_Mauritania.csv` (69,868 bytes). SHA-256:
`6da0a6b44266f448393c92a293f24f145d803ee95b6246804a0185bbfd9869f6`

[1] Oortwijn, T., Lameris, T., Zhemchuzhnikov, M., et al. (2025). *Demand-resource mismatch explains body shrinkage in a migratory shorebird* [Dataset]. Dryad. Published 4 April 2025. https://doi.org/10.5061/dryad.3n5tb2rqb